Partitioning and High-level Architecture

This lesson gives a brief overview of BigTable's architecture and its data partitioning scheme.

Table partitioning

A single instance of a BigTable implementation is known as a cluster. Each cluster can store a number of tables where each table is split into multiple **Tablets**, each around 100–200 MB in size.

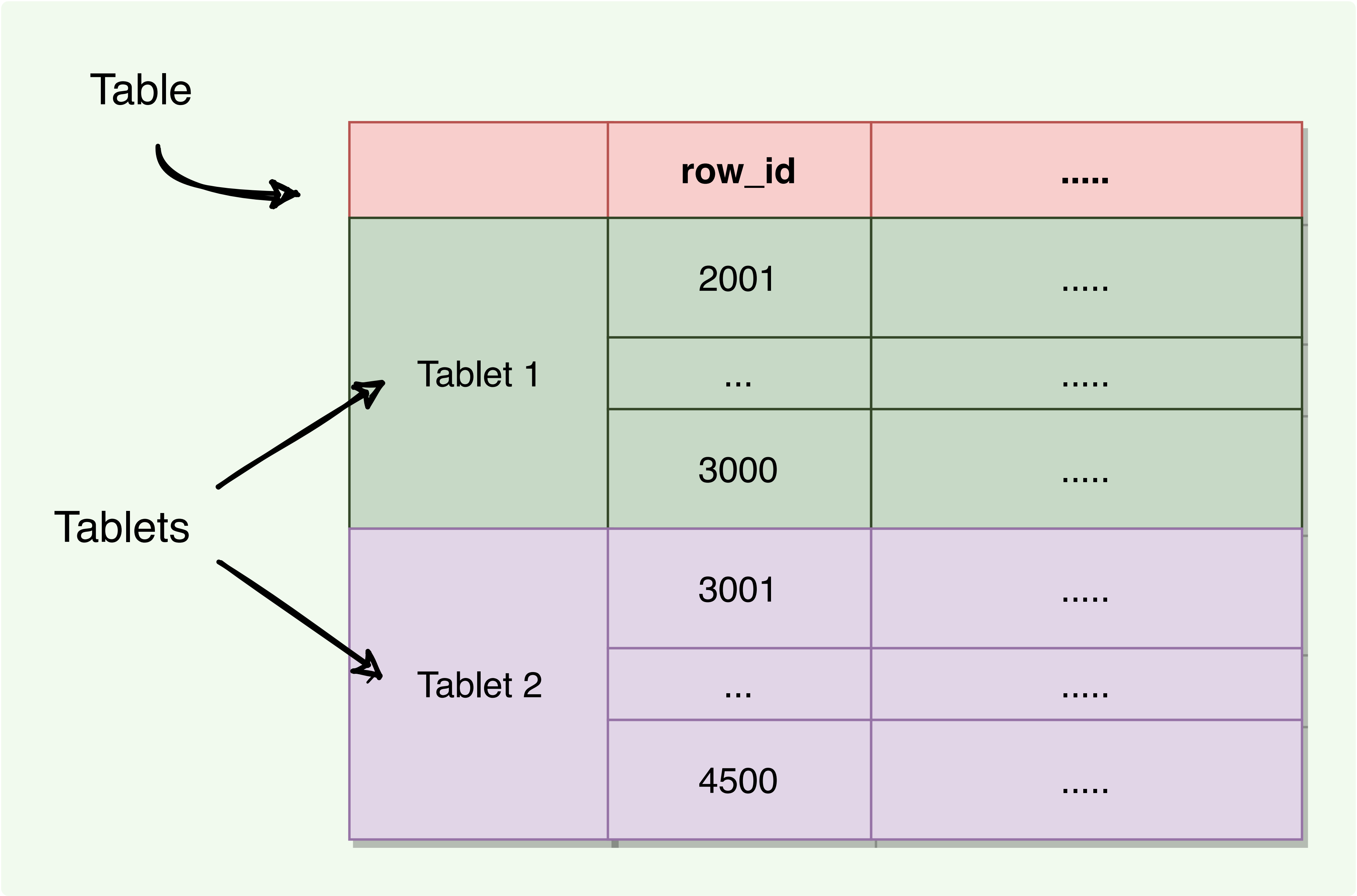


Table
Tablets
row_id
.....
Tablet 1
2001
...
3000
Tablet 2
3001
...
4500

- A Tablet holds a contiguous range of rows.
- The table is broken into Tablets at row boundaries.
- Initially, each table consists of only one Tablet. As the table grows, multiple Tablets are created. By default, a table is split at around 100 to 200 MB.
- Tablets are the unit of distribution and load balancing (more about this later).
- Since the table is sorted by row, reads of short ranges of rows are always efficient, that is to say, communicating with a small number of Tablets. This also means that selecting a row key with a high degree of locality is very important.
- Each Tablet is assigned to a **Tablet server** (discussed later), which manages all read/write requests of that Tablet.

High-level architecture

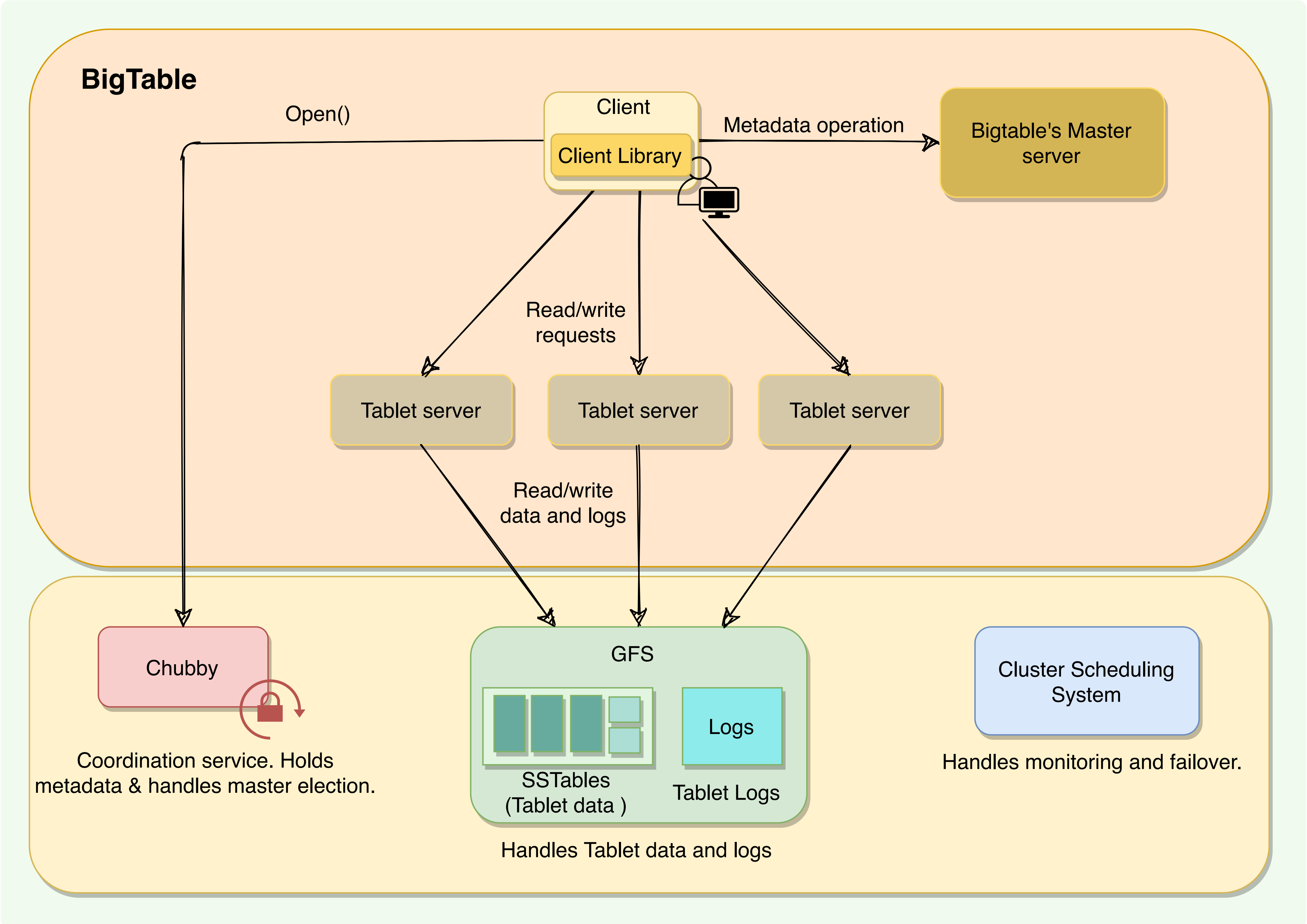
The architecture of a BigTable cluster consists of three major components:

1. **Client Library**: A library component that is linked into every client. The client talks to BigTable through this library.
2. **One master server**: Responsible for performing metadata operations and assigning Tablets to Tablet servers and managing them.
3. **Many Tablet servers**: Each Tablet server serves read and write of the data to the Tablets it is assigned.

BigTable is built on top of several other pieces from Google infrastructure:

1. **GFS**: BigTable uses the Google File System to store its data and log files.
2. **SSTable**: Google's SSTable (Sorted String Table) file format is used to store BigTable data. SSTable provides a persistent, ordered, and immutable map from keys to values (more on this later). SSTable is designed in such a way that any data access requires, at most, a single disk access.
3. **Chubby**: BigTable uses a highly available and persistent distributed lock service called Chubby to handle synchronization issues and store configuration information.
4. **Cluster Scheduling System**: Google has a cluster management system that schedules, monitors, and manages the Bigtable's cluster.

Let's understand these components one by one.


BigTable
Open()
Client
Client Library
Metadata operation
Bigtable's Master server
Read/write requests
Tablet server
Tablet server
Tablet server
Read/write data and logs
Chubby
Coordination service. Holds metadata & handles master election.
GFS
SSTables (Tablet data)
Logs
Tablet Logs
Handles Tablet data and logs
Cluster Scheduling System
Handles monitoring and failover.

High-level architecture of BigTable

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